Complete Node Fronts of Pratt Forests

Generating Polynomials and a Probabilistic Interpretation of $f_n(x)$

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Abstract

For every positive integer n, the recursively defined polynomial $f_n(x)$ admits a direct interpretation in terms of the Pratt forest of n. We first define node fronts and complete node fronts for arbitrary finite rooted trees and forests. We then associate with a rooted forest a generating polynomial in which the exponent records how many leaves are selected by a complete front. The abstract tree recursion is

$$C_T(x) = 1 + \prod_i C_{T_i}(x),$$

where T_i are the rooted subtrees below the children of the root, while disjoint union of forests corresponds to multiplication. Since the leaves of a Pratt forest are precisely the vertices labelled 2, these recursions coincide exactly with the defining recursions of the polynomials $f_n(x)$. Consequently,

$$f_n(x) = \sum_{A \in \mathrm{CF}(\mathcal{F}_n)} x^{\#\{v \in A\colon \ell(v)=2\}},$$

where $\mathcal{F}_n$ is the Pratt forest of n.

After normalization by $f_n(1)$, this polynomial becomes the probability generating function of the number of terminal 2-vertices in a uniformly random complete front. We give the associated finite probability space explicitly, derive formulas for the expectation, variance, factorial moments, and cumulants, and prove that the mean, variance, and every cumulant are completely additive arithmetic functions. For odd primes, the random variable is an explicit mixture of the zero variable and the corresponding random variable for $p-1$. Several numerical examples are included.

1 Finite rooted trees and forests

Definition 1.1 (Finite rooted tree)**.** A *finite rooted tree* T is a finite connected acyclic graph together with a distinguished vertex r_T, called its *root*. For every vertex $v \in V(T)$ there is a unique simple path from r_T to v.

A vertex u is an *ancestor* of a vertex v, written $u \preceq v$, if u lies on the unique path from r_T to v. In this case v is a *descendant* of u. A vertex with no children is called a *leaf*. The set of leaves of T is denoted by $\mathrm{Leaves}(T)$.

The ancestor relation is a partial order on $V(T)$. The root is the unique least element, and the leaves are the maximal elements.

Definition 1.2 (Finite rooted forest)**.** A *finite rooted forest* is a finite disjoint union

$$F = T_1 \,\dot{\sqcup}\, \cdots \,\dot{\sqcup}\, T_r$$

of finite rooted trees. Vertices belonging to different connected components are declared incomparable. The empty forest is denoted by $\varnothing$.

A *root-to-leaf path* in a rooted forest is a maximal chain in one of its components, beginning at the root of that component and ending at a leaf.

2 Node fronts and complete node fronts

Definition 2.1 (Node front)**.** Let F be a finite rooted forest. A subset $A \subseteq V(F)$ is called a *node front* if it is an antichain for the ancestor order: whenever $u, v \in A$ are distinct, neither $u \preceq v$ nor $v \preceq u$ holds.

Thus a front never contains both a node and one of its proper descendants.

Definition 2.2 (Complete node front)**.** A node front $A \subseteq V(F)$ is called *complete* if it meets every root-to-leaf path in F.

Since a node front is an antichain, it can meet a fixed root-to-leaf path at most once. Hence completeness is equivalent to meeting every such path exactly once.

We write

$$\mathrm{CF}(F) = \{A \subseteq V(F) : A \text{ is a complete node front of } F\}.$$

For the empty forest we use the convention

$$\mathrm{CF}(\varnothing) = \{\varnothing\}.$$

Proposition 2.3 (Equivalent descriptions)**.** *Let A be a node front in a finite rooted forest F. The following are equivalent:*

(i) *A is complete;*

(ii) *every root-to-leaf path meets A exactly once;*

(iii) *A is maximal, with respect to inclusion, among node fronts of F.*

Proof. The equivalence of (i) and (ii) follows immediately from the antichain condition: a root-to-leaf path is totally ordered, so it cannot contain two distinct vertices of A.

Assume (i). Let $v \in V(F) \setminus A$. Choose a root-to-leaf path P containing v; such a path exists because the tree is finite and every vertex has a descendant leaf. Since A is complete, P contains a vertex $a \in A$. Both a and v lie on the same chain P, so they are comparable. Therefore v cannot be added to A without destroying the antichain property. Hence A is maximal and (iii) holds.

Conversely, assume (iii) and suppose that some root-to-leaf path P does not meet A. Let λ be the terminal leaf of P. No element of A can be an ancestor of λ, since every ancestor of λ lies on P. Also λ cannot be a proper ancestor of an element of A, because λ is a leaf. Thus λ is incomparable with every element of A, and $A \cup \{\lambda\}$ is a larger node front. This contradicts maximality. Therefore every root-to-leaf path meets A, so (i) holds. $\square$

3 The generating polynomial of complete fronts

Definition 3.1 (Leaf statistic)**.** Let F be a finite rooted forest and $A \in \mathrm{CF}(F)$. Define

$$\lambda_F(A) := \#(A \cap \mathrm{Leaves}(F)).$$

Thus $\lambda_F(A)$ is the number of selected vertices of the front that are leaves of the ambient forest.

Definition 3.2 (Complete-front generating polynomial)**.** The *complete-front generating polynomial* of F is

$$\boxed{C_F(x) := \sum_{A \in \mathrm{CF}(F)} x^{\lambda_F(A)}.}$$

Because F is finite, this is a polynomial with nonnegative integer coefficients.

Its coefficients have the direct counting interpretation

$$[x^j]C_F(x) = \#\{A \in \mathrm{CF}(F) : \lambda_F(A) = j\}.$$

3.1 The recursion for a rooted tree

Let T be a rooted tree with root r. If r is not a leaf, let $T_1, \ldots, T_k$ be the rooted subtrees whose roots are the children of r.

Lemma 3.3 (Root decomposition of complete fronts)**.** *Assume that T has at least two vertices. Every complete node front A of T is of exactly one of the following two types:*

(i) $A = \{r\}$*;*

(ii) $r \notin A$*, and there are uniquely determined fronts* $A_i \in \mathrm{CF}(T_i)$ *such that*

$$A = A_1 \,\dot\sqcup \cdots \dot\sqcup\, A_k.$$

Conversely, every union of the form in (ii) *is a complete node front of T.*

Proof. Let $A \in \mathrm{CF}(T)$. If $r \in A$, then every other vertex of T is a descendant of r. Since A is an antichain, no other vertex can lie in A, and therefore $A = \{r\}$.

Now suppose $r \notin A$. For each i, put $A_i = A \cap V(T_i)$. Every root-to-leaf path in T_i extends to a root-to-leaf path in T by adjoining the root r. Since $r \notin A$ and A meets the extended path exactly once, A_i meets the original path exactly once. Moreover, A_i is an antichain because A is one. Hence $A_i \in \mathrm{CF}(T_i)$. The components T_i are disjoint, so

$$A = A_1 \,\dot\sqcup \cdots \dot\sqcup\, A_k.$$

Uniqueness is clear from $A_i = A \cap V(T_i)$.

Conversely, let $A_i \in \mathrm{CF}(T_i)$ for every i, and put $A = \dot\sqcup_i A_i$. Vertices from different T_i are incomparable, while each A_i is an antichain, so A is a node front. Every root-to-leaf path of T enters exactly one subtree T_i and is met exactly once by A_i. Thus A is complete. □

Theorem 3.4 (Tree recursion)**.** *Let T be a finite rooted tree.*

(i) *If T consists of a single vertex, then*

$$C_T(x) = x.$$

(ii) *If the root of T has rooted child-subtrees $T_1, \dots, T_k$, then*

$$\boxed{C_T(x) = 1 + \prod_{i=1}^{k} C_{T_i}(x).}$$

Proof. If T consists of one vertex r, then r is a leaf and the unique complete front is $\{r\}$. Its leaf statistic is 1, so $C_T(x) = x$.

Now assume that T is nontrivial. By the preceding lemma, the complete fronts split into two disjoint classes. The first class consists only of $\{r\}$. Since the root is not a leaf, its contribution is $x^0 = 1$.

For the second class, choose independently a complete front $A_i \in \mathrm{CF}(T_i)$ for each i and take their union. Since the leaves of T lying in T_i are precisely the leaves of T_i, the leaf statistic is additive:

$$\lambda_T(A_1 \dot\sqcup \cdots \dot\sqcup A_k) = \sum_{i=1}^{k} \lambda_{T_i}(A_i).$$

Therefore

$$\begin{aligned}\sum_{A_1 \in \mathrm{CF}(T_1)} \cdots \sum_{A_k \in \mathrm{CF}(T_k)} x^{\sum_i \lambda_{T_i}(A_i)} &= \prod_{i=1}^{k} \left(\sum_{A_i \in \mathrm{CF}(T_i)} x^{\lambda_{T_i}(A_i)} \right) \\ &= \prod_{i=1}^{k} C_{T_i}(x).\end{aligned}$$

Adding the contribution of the root front gives the claimed formula. □

3.2 The recursion for a rooted forest

Theorem 3.5 (Forest multiplicativity). *If*

$$F = T_1 \dot\sqcup \cdots \dot\sqcup T_r,$$

then

$$\boxed{C_F(x) = \prod_{i=1}^{r} C_{T_i}(x).}$$

For the empty forest, $C_\varnothing(x) = 1$.

Proof. A root-to-leaf path of F lies in exactly one component. Hence a complete front of F is uniquely a disjoint union

$$A = A_1 \dot\sqcup \cdots \dot\sqcup A_r, \qquad A_i \in \mathrm{CF}(T_i).$$

The leaf statistic is additive across components:

$$\lambda_F(A) = \sum_{i=1}^{r} \lambda_{T_i}(A_i).$$

Thus the same product-of-sums argument as above gives

$$C_F(x) = \prod_{i=1}^{r} C_{T_i}(x).$$

For the empty forest there is exactly one complete front, namely $\varnothing$, and its leaf statistic is 0. Hence $C_\varnothing(x) = 1$. □

4 Pratt trees, Pratt forests, and the polynomials $f_n(x)$

Definition 4.1 (Pratt tree)**.** For every prime p, define a finite rooted tree T_p recursively as follows.

- T_2 consists of one vertex labelled 2.
- If p is odd and
$$p-1=\prod_{q\mid p-1} q^{v_q(p-1)},$$
then T_p has a root labelled p, and for every prime divisor q of $p-1$ it has $v_q(p-1)$ children, each carrying a copy of the rooted tree T_q below it.

The recursion is well founded because every prime divisor q of $p-1$ satisfies $q<p$.

Definition 4.2 (Pratt forest)**.** For
$$n=\prod_p p^{v_p(n)},$$
the Pratt forest of n is
$$\boxed{\mathcal{F}_n := \dot\sqcup_p v_p(n)\, T_p,}$$
where $v_p(n)\,T_p$ denotes the disjoint union of $v_p(n)$ copies of T_p. In particular, $\mathcal{F}_1=\varnothing$.

Lemma 4.3 (Leaves of a Pratt forest)**.** *The leaves of every Pratt tree, and hence of every Pratt forest, are exactly the vertices labelled* 2*.*

Proof. The unique vertex of T_2 is a leaf. Let p be an odd prime. Since $p-1\geq 2$, the integer $p-1$ has at least one prime divisor, so the root of T_p has at least one child and is not a leaf. The same argument applies to every odd-labelled vertex occurring recursively inside a Pratt tree. Thus no odd-labelled vertex is a leaf. Every vertex labelled 2 is the root of a copy of T_2, so it has no children and is a leaf. □

Definition 4.4 (The recursive polynomials)**.** Define polynomials $f_n(x)\in\mathbb{Z}[x]$ by
$$f_1(x)=1,\qquad f_2(x)=x.$$
For every odd prime p, define
$$\boxed{f_p(x)=1+\prod_{q\mid p-1} f_q(x)^{v_q(p-1)}.}$$
For an arbitrary positive integer
$$n=\prod_p p^{v_p(n)},$$
define
$$\boxed{f_n(x)=\prod_p f_p(x)^{v_p(n)}.}$$

Theorem 4.5 (Complete-front interpretation of f_n)**.** *For every positive integer n,*
$$\boxed{f_n(x)=C_{\mathcal{F}_n}(x).}$$

Equivalently,

$$\boxed{f_n(x) = \sum_{A \in \mathrm{CF}(\mathcal{F}_n)} x^{\#\{v \in A:\, \ell(v)=2\}},}$$

where $\ell(v)$ denotes the prime label of the vertex v. Consequently,

$$[x^j]f_n(x) = \#\left\{A \in \mathrm{CF}(\mathcal{F}_n) : \#\{v \in A : \ell(v) = 2\} = j\right\}.$$

Proof. We first prove the statement for prime indices. The proof is by induction on the prime p.

For $p = 2$, the tree T_2 consists of one vertex. By the one-vertex case of the abstract tree recursion,

$$C_{T_2}(x) = x = f_2(x).$$

Now let p be an odd prime and assume that $C_{T_q}(x) = f_q(x)$ for every prime $q < p$. The child-subtrees of the root of T_p consist, for each prime $q \mid p-1$, of exactly $v_q(p-1)$ copies of T_q. The tree recursion therefore gives

$$C_{T_p}(x) = 1 + \prod_{q \mid p-1} C_{T_q}(x)^{v_q(p-1)}.$$

Every prime divisor q of $p-1$ is smaller than p, so the induction hypothesis applies and yields

$$C_{T_p}(x) = 1 + \prod_{q \mid p-1} f_q(x)^{v_q(p-1)} = f_p(x).$$

Thus $C_{T_p}(x) = f_p(x)$ for every prime p.

Finally, let $n = \prod_p p^{v_p(n)}$. By definition,

$$\mathcal{F}_n = \dot\sqcup_p v_p(n)\, T_p.$$

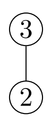
Forest multiplicativity and the prime case imply

$$C_{\mathcal{F}_n}(x) = \prod_p C_{T_p}(x)^{v_p(n)} = \prod_p f_p(x)^{v_p(n)} = f_n(x).$$

The equivalent sum formula follows because the leaves of $\mathcal{F}_n$ are precisely the vertices labelled 2. $\square$

5 Examples

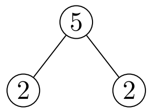
Example 5.1 (The prime 3)**.** The Pratt tree is



There are two complete fronts: $\{3\}$ and $\{2\}$. Their leaf statistics are 0 and 1, respectively. Hence

$$C_{T_3}(x) = 1 + x = f_3(x).$$

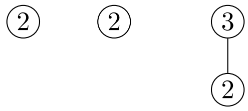
Example 5.2 (The prime 5)**.** Since $5 - 1 = 2^2$, the Pratt tree has two terminal children:



There are two complete fronts: the root front $\{5\}$, of leaf statistic 0, and the leaf front consisting of both 2-vertices, of leaf statistic 2. Therefore

$$C_{T_5}(x) = 1 + x^2 = f_5(x).$$

Example 5.3 (The integer 12)**.** Since $12 = 2^2 \cdot 3$, its Pratt forest consists of two isolated copies of T_2 and one copy of T_3:



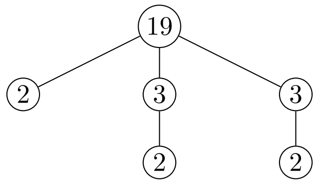
The two isolated 2-vertices must belong to every complete front. In the component T_3 one may choose either the root 3 or its leaf 2. Thus there is one complete front with two selected leaves and one with three selected leaves. Consequently,

$$C_{\mathcal{F}_{12}}(x) = x^2 + x^3.$$

The polynomial recursion gives the same result:

$$f_{12}(x) = f_2(x)^2 f_3(x) = x^2(1 + x) = x^2 + x^3.$$

Example 5.4 (The prime 19)**.** Since $19 - 1 = 18 = 2 \cdot 3^2$, the root of T_{19} has one child of type T_2 and two children of type T_3:



The root front contributes 1. If the root is not chosen, the terminal T_2 contributes x, while each copy of T_3 contributes $1 + x$. Therefore

$$\begin{aligned} C_{T_{19}}(x) &= 1 + x(1 + x)^2 \\ &= 1 + x + 2x^2 + x^3. \end{aligned}$$

Thus the tree has one complete front selecting no leaves, one selecting one leaf, two selecting two leaves, and one selecting three leaves. In particular,

$$C_{T_{19}}(1) = 5.$$

6 Immediate consequences

Corollary 6.1 (Number of complete fronts)**.** *For every positive integer n,*

$$\boxed{f_n(1) = \#\,\mathrm{CF}(\mathcal{F}_n).}$$

Proof. At $x = 1$, every monomial in the complete-front generating polynomial has weight 1. Hence the sum counts all complete fronts. $\square$

Corollary 6.2 (Fronts without leaves)**.** *For every positive integer n,*

$$\boxed{f_n(0) = \begin{cases} 0, & 2 \mid n, \\ 1, & 2 \nmid n. \end{cases}}$$

Proof. If $2 \mid n$, then $\mathcal{F}_n$ contains at least one component equal to T_2. Its unique vertex is a leaf and must lie in every complete front, so no complete front has leaf statistic 0.

If n is odd, every component of $\mathcal{F}_n$ has an odd prime at its root. Choosing the root of every component gives one leaf-free complete front. It is the only one: if a front does not choose the root of a component, it must descend into that component, and every complete descent eventually selects at least one leaf labelled 2. □

Corollary 6.3 (Degree and leading coefficient)**.** *Let $m_2(n)$ denote the number of vertices labelled 2 in the Pratt forest of n. Then*

$$\boxed{\deg f_n = m_2(n),}$$

and $f_n(x)$ is monic.

Proof. The set of all leaves of a finite rooted forest is a complete node front. In a Pratt forest these leaves are exactly the 2-labelled vertices, so this front has leaf statistic $m_2(n)$. No complete front can contain more leaves than the total number of leaves. Hence the maximal exponent of x in $f_n(x)$ is $m_2(n)$.

There is exactly one complete front containing all leaves, namely the leaf set itself. Therefore the coefficient of $x^{m_2(n)}$ is 1. □

Corollary 6.4 (Weighted front formula for the integer n)**.** *For every positive integer n,*

$$\boxed{n = \sum_{A \in \mathrm{CF}(\mathcal{F}_n)} 2^{\#\{v \in A : \ell(v) = 2\}}.}$$

Proof. By the main theorem, the right-hand side is $f_n(2)$. It remains to verify that $f_n(2) = n$. This follows by induction from the defining recursion. We have $f_1(2) = 1$ and $f_2(2) = 2$. If p is an odd prime and the identity is known for every prime divisor of $p - 1$, then

$$\begin{aligned} f_p(2) &= 1 + \prod_{q \mid p-1} f_q(2)^{v_q(p-1)} \\ &= 1 + \prod_{q \mid p-1} q^{v_q(p-1)} \\ &= 1 + (p - 1) = p. \end{aligned}$$

For arbitrary $n = \prod_p p^{v_p(n)}$, multiplicativity gives

$$f_n(2) = \prod_p f_p(2)^{v_p(n)} = \prod_p p^{v_p(n)} = n.$$

□

7 A probability space of complete fronts

The complete-front interpretation defines a canonical finite probability model. Fix $n \geq 1$. We choose one complete node front of $\mathcal{F}_n$ uniformly at random.

Definition 7.1 (The complete-front probability space)**.** Define

$$\Omega_n := \mathrm{CF}(\mathcal{F}_n), \qquad \Sigma_n := 2^{\Omega_n}.$$

Since Ω_n is finite, define the uniform probability measure $\mathbb{P}_n$ by

$$\boxed{\mathbb{P}_n(E) = \frac{\#E}{\#\Omega_n} \qquad (E \subseteq \Omega_n).}$$

In particular, every individual complete front $A \in \Omega_n$ has probability

$$\mathbb{P}_n(\{A\}) = \frac{1}{\#\,\mathrm{CF}(\mathcal{F}_n)} = \frac{1}{f_n(1)}.$$

Definition 7.2 (Random terminal-leaf count)**.** Define the random variable

$$X_n : \Omega_n \longrightarrow \mathbb{N}_0$$

by

$$\boxed{X_n(A) := \lambda_{\mathcal{F}_n}(A) = \#\{v \in A : \ell(v) = 2\}.}$$

Thus X_n counts the terminal 2-vertices selected by a uniformly random complete front of the Pratt forest of n.

Write

$$f_n(x) = \sum_{j \geq 0} c_j(n) x^j.$$

By the complete-front theorem,

$$c_j(n) = \#\{A \in \mathrm{CF}(\mathcal{F}_n) : X_n(A) = j\}.$$

Hence

$$\mathbb{P}_n(X_n = j) = \frac{c_j(n)}{f_n(1)}.$$

Theorem 7.3 (Normalized probability generating function)**.** *The probability generating function of X_n is*

$$\boxed{G_n(s) := \mathbb{E}_n[s^{X_n}] = \frac{f_n(s)}{f_n(1)}.}$$

Equivalently,

$$\mathbb{P}_n(X_n = j) = \frac{[s^j] f_n(s)}{f_n(1)}.$$

Proof. Uniformity gives

$$\begin{aligned} \mathbb{E}_n[s^{X_n}] &= \sum_{A \in \Omega_n} s^{X_n(A)} \mathbb{P}_n(\{A\}) \\ &= \frac{1}{\#\Omega_n} \sum_{A \in \mathrm{CF}(\mathcal{F}_n)} s^{X_n(A)}. \end{aligned}$$

The numerator is $f_n(s)$ by the complete-front interpretation, while $\#\Omega_n = f_n(1)$. This proves the formula. $\square$

8 Expectation and factorial moments

The first derivative has an immediate counting meaning.

Proposition 8.1 (Total selected-leaf count)**.** *One has*

$$\boxed{f_n'(1) = \sum_{A \in \mathrm{CF}(\mathcal{F}_n)} X_n(A).}$$

Thus $f_n'(1)$ is the total number of selected terminal 2-vertices, counted over all complete fronts of $\mathcal{F}_n$.

Proof. Differentiating

$$f_n(s) = \sum_{j \ge 0} c_j(n) s^j$$

gives

$$f_n'(s) = \sum_{j \ge 1} j\, c_j(n) s^{j-1}.$$

At $s = 1$,

$$f_n'(1) = \sum_{j \ge 1} j\, c_j(n).$$

For each j, the number $c_j(n)$ counts the complete fronts containing exactly j selected terminal vertices. Hence $jc_j(n)$ is their total contribution to the selected-leaf count. Summing over j proves the result. $\square$

Corollary 8.2 (Expected number of selected terminal vertices)**.** *Let*

$$\mu(n) := \mathbb{E}_n[X_n].$$

Then

$$\boxed{\mu(n) = \frac{f_n'(1)}{f_n(1)} = \frac{d}{ds} \log f_n(s)\Big|_{s=1}.}$$

Proof. The derivative of the probability generating function at 1 equals the expectation:

$$G_n'(1) = \sum_j j\, \mathbb{P}_n(X_n = j) = \mathbb{E}_n[X_n].$$

Since $G_n(s) = f_n(s)/f_n(1)$, the first formula follows. The second is the definition of the logarithmic derivative. $\square$

For $r \ge 0$, write

$$(x)_r = x(x-1)\cdots(x-r+1), \qquad (x)_0 = 1,$$

for the falling factorial.

Proposition 8.3 (Factorial moments)**.** *For every integer $r \ge 0$,*

$$\boxed{\mathbb{E}_n[(X_n)_r] = \frac{f_n^{(r)}(1)}{f_n(1)}.}$$

Proof. Differentiating the probability generating function r times gives

$$G_n^{(r)}(s) = \sum_j (j)_r \, \mathbb{P}_n(X_n = j) s^{j-r}.$$

Evaluation at $s = 1$ yields $G_n^{(r)}(1) = \mathbb{E}_n[(X_n)_r]$. Since $f_n(1)$ is constant with respect to s, one has

$$G_n^{(r)}(1) = \frac{f_n^{(r)}(1)}{f_n(1)}.$$

□

9 Products of forests and convolution of distributions

The multiplicativity $f_{ab} = f_a f_b$ has an exact probabilistic interpretation.

Theorem 9.1 (Independent-sum decomposition)**.** *For all positive integers a, b, a uniformly random complete front of*

$$\mathcal{F}_{ab} = \mathcal{F}_a \,\dot\sqcup\, \mathcal{F}_b$$

corresponds to an independent uniformly random pair of complete fronts of $\mathcal{F}_a$ and $\mathcal{F}_b$. Under this correspondence,

$$\boxed{X_{ab} = X_a + X_b.}$$

Consequently,

$$\boxed{G_{ab}(s) = G_a(s) G_b(s).}$$

Here X_a and X_b on the right are taken independently.

Proof. Every complete front of the disjoint union $\mathcal{F}_a \,\dot\sqcup\, \mathcal{F}_b$ has a unique representation

$$A = A_a \,\dot\sqcup\, A_b, \qquad A_a \in \mathrm{CF}(\mathcal{F}_a), \quad A_b \in \mathrm{CF}(\mathcal{F}_b).$$

Thus

$$\mathrm{CF}(\mathcal{F}_{ab}) \cong \mathrm{CF}(\mathcal{F}_a) \times \mathrm{CF}(\mathcal{F}_b).$$

Because

$$\#\,\mathrm{CF}(\mathcal{F}_{ab}) = \#\,\mathrm{CF}(\mathcal{F}_a)\,\#\,\mathrm{CF}(\mathcal{F}_b),$$

the uniform measure on the left corresponds exactly to the product of the two uniform measures on the right. Hence the two component fronts are independent. The selected terminal vertices in disjoint components add, so

$$X_{ab}(A_a \,\dot\sqcup\, A_b) = X_a(A_a) + X_b(A_b).$$

Taking probability generating functions proves the product formula. □

Corollary 9.2 (Complete additivity of the expectation)**.** *For all positive integers a, b,*

$$\boxed{\mu(ab) = \mu(a) + \mu(b).}$$

In particular,

$$\mu(1) = 0, \qquad \mu(n^k) = k\mu(n),$$

and if $n = \prod_p p^{v_p(n)}$, then

$$\boxed{\mu(n) = \sum_p v_p(n)\mu(p).}$$

Proof. Expectations add for independent sums. The remaining identities follow by repeated application to powers and to the prime factorization of n. □

10 Variance and cumulants

Theorem 10.1 (Variance formula)**.** *Let*

$$\sigma^2(n) := \mathrm{Var}(X_n).$$

Then

$$\boxed{\sigma^2(n) = \frac{f_n''(1)}{f_n(1)} + \frac{f_n'(1)}{f_n(1)} - \left(\frac{f_n'(1)}{f_n(1)}\right)^2.}$$

Equivalently,

$$\sigma^2(n) = \frac{f_n''(1)}{f_n(1)} + \mu(n) - \mu(n)^2.$$

Proof. The second factorial moment is

$$\mathbb{E}_n[X_n(X_n - 1)] = G_n''(1) = \frac{f_n''(1)}{f_n(1)}.$$

Since $X_n^2 = X_n(X_n - 1) + X_n$, one has

$$\mathbb{E}_n[X_n^2] = \frac{f_n''(1)}{f_n(1)} + \mu(n).$$

Subtracting $\mu(n)^2$ gives the variance formula. □

Corollary 10.2 (Complete additivity of the variance)**.** *For all positive integers a, b,*

$$\boxed{\sigma^2(ab) = \sigma^2(a) + \sigma^2(b).}$$

Consequently,

$$\sigma^2(n) = \sum_p v_p(n)\sigma^2(p).$$

Proof. By the independent-sum theorem, X_{ab} is distributed as the sum of independent copies of X_a and X_b. Variances of independent sums add. □

The same phenomenon holds at every order. Define the moment generating function and cumulant generating function by

$$M_n(t) := \mathbb{E}_n[e^{tX_n}] = G_n(e^t) = \frac{f_n(e^t)}{f_n(1)},$$

$$K_n(t) := \log M_n(t) = \log f_n(e^t) - \log f_n(1).$$

The r-th cumulant is

$$\kappa_r(n) := K_n^{(r)}(0).$$

Theorem 10.3 (Complete additivity of all cumulants)**.** *For every $r \geq 1$ and all positive integers a, b,*

$$\boxed{\kappa_r(ab) = \kappa_r(a) + \kappa_r(b).}$$

In particular,

$$\kappa_1(n) = \mu(n), \qquad \kappa_2(n) = \sigma^2(n).$$

No coprimality condition on a and b is required.

Proof. The independent-sum theorem gives

$$M_{ab}(t) = M_a(t)M_b(t).$$

Taking logarithms,

$$K_{ab}(t) = K_a(t) + K_b(t).$$

Differentiating r times at $t = 0$ proves the claim. $\square$

11 Recursive formulas at primes

Define

$$A(n) := f_n(1), \qquad B(n) := f_n'(1), \qquad C(n) := f_n''(1).$$

Thus $A(n)$ is the number of complete fronts, $B(n)$ is the total selected-leaf count, and $C(n)$ is the total second falling-factorial count.

Theorem 11.1 (Derivative recursions)**.** *The following identities hold.*

(i) *At the terminal prime,*

$$A(2) = 1, \qquad B(2) = 1, \qquad C(2) = 0.$$

(ii) *For every odd prime p,*

$$\boxed{A(p) = A(p-1) + 1, \quad B(p) = B(p-1), \quad C(p) = C(p-1).}$$

(iii) *For all positive integers a, b,*

$$\boxed{A(ab) = A(a)A(b),}$$

$$\boxed{B(ab) = B(a)A(b) + A(a)B(b),}$$

$$\boxed{C(ab) = C(a)A(b) + 2B(a)B(b) + A(a)C(b).}$$

Proof. The first statement follows from $f_2(s) = s$. For an odd prime p,

$$f_p(s) = 1 + f_{p-1}(s).$$

Evaluation at 1 gives the formula for A, while one or two differentiations eliminate the added constant and give the formulas for B and C.

For products, use $f_{ab} = f_a f_b$. Evaluation at 1 gives the formula for A. The first product rule gives the formula for B, and the second product rule

$$(f_a f_b)'' = f_a'' f_b + 2f_a' f_b' + f_a f_b''$$

gives the formula for C after evaluation at 1. $\square$

Corollary 11.2 (Prime attenuation formulas)**.** *Let p be an odd prime and put*

$$w_p := \frac{A(p-1)}{A(p-1)+1}.$$

Then

$$\boxed{\mu(p) = w_p \mu(p-1)}$$

and

$$\boxed{\sigma^2(p) = w_p \sigma^2(p-1) + w_p(1-w_p)\mu(p-1)^2.}$$

Proof. The complete fronts of T_p consist of the single root front, containing zero leaves, and the $A(p-1)$ expanded fronts naturally identified with the complete fronts of $\mathcal{F}_{p-1}$. Hence

$$X_p \overset{d}{=} \begin{cases} 0, & \text{with probability } 1-w_p, \\ X_{p-1}, & \text{with probability } w_p. \end{cases}$$

The expectation is therefore $w_p\mu(p-1)$.

For the variance, the second raw moment is multiplied by w_p:

$$\mathbb{E}[X_p^2] = w_p\mathbb{E}[X_{p-1}^2].$$

Thus

$$\begin{aligned} \operatorname{Var}(X_p) &= w_p\big(\sigma^2(p-1) + \mu(p-1)^2\big) - w_p^2\mu(p-1)^2 \\ &= w_p\sigma^2(p-1) + w_p(1-w_p)\mu(p-1)^2. \end{aligned}$$

□

The factor w_p has a direct interpretation: it is the proportion of complete fronts of T_p that expand below the root. The remaining proportion $1-w_p$ consists of the single root front.

12 Numerical examples

Example 12.1 (The prime 7)**.** One has

$$f_7(s) = 1 + s + s^2, \qquad G_7(s) = \frac{1+s+s^2}{3}.$$

Thus X_7 takes the values $0, 1, 2$, each with probability $1/3$. Therefore

$$\mu(7) = \frac{0+1+2}{3} = 1,$$

while

$$\sigma^2(7) = \frac{0^2+1^2+2^2}{3} - 1^2 = \frac{2}{3}.$$

The same mean is obtained recursively:

$$\mu(2) = 1, \qquad \mu(3) = \frac{1}{2}, \qquad \mu(6) = \mu(2) + \mu(3) = \frac{3}{2},$$

and since $A(6) = 2$,

$$\mu(7) = \frac{A(6)}{A(6)+1}\mu(6) = \frac{2}{3}\cdot\frac{3}{2} = 1.$$

Example 12.2 (The prime 19)**.** The complete-front polynomial is

$$f_{19}(s) = 1 + s + 2s^2 + s^3.$$

Hence

$$G_{19}(s) = \frac{1+s+2s^2+s^3}{5},$$

and the probabilities for $X_{19} = 0, 1, 2, 3$ are respectively

$$\frac{1}{5}, \quad \frac{1}{5}, \quad \frac{2}{5}, \quad \frac{1}{5}.$$

Since

$$f'_{19}(1) = 8, \qquad f''_{19}(1) = 10,$$

one obtains

$$\boxed{\mu(19) = \frac{8}{5}}$$

and

$$\boxed{\sigma^2(19) = \frac{10}{5} + \frac{8}{5} - \left(\frac{8}{5}\right)^2 = \frac{26}{25}.}$$

Example 12.3 (The prime 43)**.** Since $43 - 1 = 42 = 2 \cdot 3 \cdot 7$,

$$f_{43}(s) = 1 + f_2(s) f_3(s) f_7(s) = 1 + s + 2s^2 + 2s^3 + s^4.$$

Thus

$$G_{43}(s) = \frac{1 + s + 2s^2 + 2s^3 + s^4}{7}.$$

A direct calculation gives

$$f'_{43}(1) = 15, \qquad f''_{43}(1) = 28.$$

Therefore

$$\boxed{\mu(43) = \frac{15}{7}}$$

and

$$\boxed{\sigma^2(43) = \frac{28}{7} + \frac{15}{7} - \left(\frac{15}{7}\right)^2 = \frac{76}{49}.}$$

Example 12.4 (Fermat-prime form)**.** Suppose $p = 2^k + 1$ is prime. Then

$$f_{p-1}(s) = f_{2^k}(s) = s^k$$

and therefore

$$\boxed{f_p(s) = 1 + s^k, \qquad G_p(s) = \frac{1 + s^k}{2}.}$$

Thus

$$X_p = \begin{cases} 0, & \text{with probability } 1/2, \\ k, & \text{with probability } 1/2. \end{cases}$$

Consequently,

$$\boxed{\mu(p) = \frac{k}{2}, \qquad \sigma^2(p) = \frac{k^2}{4}.}$$

The factor 1/2 in the mean is the strongest possible root attenuation: exactly one of the two complete fronts stops at the root, while the other reaches all k terminal vertices.

13 Interpretive remarks

Remark 13.1 (Front width, not tree depth). The random variable X_n measures a cut statistic: the number of terminal 2-vertices appearing in a uniformly random complete front. It does not equal the height of the Pratt tree, the total number of vertices in a certificate, or the cost of a primality verification. It may nevertheless serve as a new average-width statistic for Pratt forests.

Remark 13.2 (Two distinct probabilistic questions). The probability space developed here keeps n fixed and chooses a random complete front. A different problem chooses n randomly from an interval and studies the additive arithmetic functions $\mu(n)$, $\sigma^2(n)$, or $\kappa_r(n)$. An Erdős–Kac-type theorem would concern this second layer of randomness and does not follow automatically from the complete-front probability model.

Remark 13.3 (Possible normal approximation along large forests). If a sequence of Pratt forests decomposes into many components and no single component dominates the total variance, then X_n is a sum of many independent component variables. Classical central-limit criteria can therefore be tested along such sequences. This is a consequence of the exact independent-sum decomposition, but a general limit theorem requires additional hypotheses.